

AEGIS — Autonomous Self-Healing Cloud Infrastructure

Research Papers, AWS Resources & YouTube Learning Plan

1. Core Research Papers

AIOpsLab: A Holistic Framework to Evaluate AI Agents for Enabling Autonomous Clouds

Priority: 10/10 — Start here. Covers fault localization, root-cause analysis, fault injection, telemetry, incident lifecycle, and evaluation of autonomous cloud agents.

Link: <https://arxiv.org/abs/2501.06706>

Building AI Agents for Autonomous Clouds: Challenges and Design Principles

Priority: 9/10 — Provides design principles and requirements for AI agents operating cloud services and discusses self-healing cloud architectures.

Link: <https://doi.org/10.1145/3698038.3698525>

CROSS — Cloud-Native Automated Remediation and Self-Healing

Priority: 8/10 — Useful for understanding observability-driven automated remediation and recovery policies.

Link: <https://link.springer.com/article/10.1186/s42400-026-00549-8>

Distributed Fault Root Cause Localization and Self-Healing Strategy Generation Based on Causal Inference

Priority: 9/10 — Study this after the MVP. Focuses on causal reasoning for root-cause localization and safer self-healing decisions.

Link: <https://doi.org/10.1016/j.procs.2026.04.270>

AI-driven Self-Healing Across the Edge-Cloud Continuum

Priority: 7/10 — Systematic review covering AI-driven self-healing, MAPE-K, reinforcement learning, generative AI, decentralized systems, and human-in-the-loop safety.

Link: <https://www.sciencedirect.com/science/article/pii/S0950584926002211>

2. AWS Practical Resources

AWS Self-Healing Mechanism

Use as the Day-1 baseline for CloudWatch/EventBridge/Lambda-based automated recovery.

Link: <https://builder.aws.com/content/2zSXztKZo3xjMPtjvyJEgk1HQBt/self-healing-mechanism-for-a-lightweight-website>

AWS EventBridge Tutorial

Official tutorial showing how EventBridge rules can route events to Lambda functions.

Link: <https://docs.aws.amazon.com/eventbridge/latest/userguide/eb-tutorial-get-started.html>

Microsoft AIOpsLab GitHub

Open-source framework for deploying microservices, injecting faults, exporting telemetry, and evaluating autonomous AIOps agents.

Link: <https://github.com/microsoft/AIOpsLab>

3. YouTube Videos

AWS Lambda Triggering with EventBridge & CloudWatch Rules — Step-by-Step

Covers EventBridge rules, Lambda triggers, and CloudWatch logs.

Link: <https://www.youtube.com/watch?v=AsgsU7RLRf4>

AWS: Trigger Lambda Function Using CloudWatch Event / EventBridge Rule

Short practical demonstration of creating a Lambda function and triggering it with an EventBridge rule.

Link: <https://www.youtube.com/watch?v=1OUuBNJFIT4>

YouTube Search — AWS CloudWatch + Lambda + EventBridge

Use this to find longer hands-on tutorials as needed.

Link:
https://www.youtube.com/results?search_query=AWS+CloudWatch+Lambda+EventBridge+tutorial

YouTube Search — AIOps / Autonomous Cloud / Self-Healing

Useful for understanding the broader research direction.

Link:
https://www.youtube.com/results?search_query=AIOps+autonomous+cloud+self+healing+root+cause+analysis

4. Recommended Study Order

1. Read AIOpsLab first and understand Detection → Localization → Root Cause Analysis → Mitigation.
2. Read the AWS self-healing example to understand the basic CloudWatch → EventBridge → Lambda workflow.
3. Watch the EventBridge + Lambda video and reproduce the trigger workflow.
4. Build the Day-1 MVP: deliberately fail a service, detect it, trigger Lambda, recover it, and verify recovery.
5. Read Building AI Agents for Autonomous Clouds before adding AI reasoning.
6. Read CROSS for automated remediation concepts.
7. Read the causal-inference paper when you are ready to make root-cause analysis more advanced.
8. Use AIOpsLab's GitHub implementation as a research reference and benchmark inspiration, not as a project to copy.

5. Target AEGIS Architecture

Observe → Detect → Diagnose → Decide → Remediate → Verify → Resolve/Escalate

Recommended AWS stack: CloudWatch, EventBridge, Lambda, SQS, Step Functions, DynamoDB, CloudTrail, IAM, and optionally Amazon Bedrock for assisted diagnosis/explanation.